



Implementation & Feasibility Plan

How Medreach goes from a working prototype to a state-wide service: phases, integrations, people, cost drivers and risks.

☆ MPOne Idea & Innovation Hackathon 2026 🏠 Challenge 5 · AI for Public Services · Healthcare

👤 Team Leader: Prakhar Porwal 🌐 medreach-wtne.onrender.com 📄 github.com/prakharporwal518-collab/Medreach

1 Why this is realistic: it already works

Live

Deployed at medreach-wtne.onrender.com

2

Portals: citizen & hospital staff

16

Bhopal-region hospitals modelled

51

Automated tests passing



Technical feasibility

Built on proven open technology (Node.js, PWA, OpenStreetMap). Runs on one small server; AI is optional and has an offline fallback.



Operational feasibility

Hospitals already have emergency desks. Staff only confirm or update figures; one click re-verifies. Roles match existing posts.



Legal & policy fit

Designed around consent, minimum data and retention limits in line with the DPDP Act 2023; ABDM-ready for ABHA records.

2 Phased rollout plan

Phase	Duration	Activities	Exit criteria
0 · Prototype Done	Completed	Citizen + hospital portals, AI triage, matching, acceptance, live maps, "I'm alone", privacy controls, deployment.	Working demo, tests passing.
1 · Pilot Bhopal	~3 months	Onboard a group of government and private hospitals; appoint a nodal officer per hospital; train ER desks; clinical review of triage rules; SMS gateway for OTP.	Hospitals keep data fresh; KPIs measured; staff feedback positive.
2 · Integrate	~3–6 months	Connect NHA Health Facility Registry, hospital HMIS feeds, 108 control room (CAD) hand-off, ABDM consent-based records; persistent encrypted database.	Live data without manual entry where HMIS exists.
3 · Scale MP state	~6–12 months	District-by-district rollout, state command dashboard, IVR / missed-call channel for feature phones, more languages.	State-wide coverage; public launch.

Durations are planning estimates and depend on hospital onboarding and integration approvals.

Sustainability: Medreach can run as a Govt. of MP digital public service (like 108), hosted on state cloud, with hospitals onboarded through the health department and Ayushman empanelment.

3 Who does what

Role	Responsibility in Medreach
Hospital Nodal Officer	Owns the hospital's data, manages staff access, reviews the audit log.
Emergency Desk Officer	Accepts / declines referrals, sets ER open / busy / diverting, confirms figures.
Resource Manager	Updates beds, specialists on duty and equipment status.
State Health Admin	Read-only view of all hospitals, audit logs and performance.

4 Integrations needed for live mode



NHA Health Facility Registry

Official list and location of hospitals and their services, instead of manual entry.



Hospital HMIS

Beds, ICU and equipment status synced automatically where a hospital system exists.



108 control room (CAD)

Ambulance request handed to the authorised control room. Medreach does not control 108 itself.



ABDM / ABHA

Consent-based access to health records instead of OCR only.



SMS gateway

Staff OTP and trusted-contact alerts (simulated in the prototype).



OpenStreetMap + routing

Free maps with no API keys, plus road routing with a straight-line fallback.

5 Cost drivers (low by design)

What keeps cost low

- Open-source stack; no licence fees.
- Free OpenStreetMap maps with no per-view API charges.
- PWA: one app for Android, iPhone and desktop, with no app-store builds needed.
- AI is optional; the offline rules engine costs nothing to run.

Main recurring costs

- Cloud hosting and a managed database.
- SMS for OTP and trusted-contact alerts.
- AI usage (per request, only when enabled).
- Hospital onboarding, training and support team.

6 Risks and how we handle them

Risk	Mitigation already built or planned
Hospitals don't update data	Freshness shown on every figure; stale data is marked unverified and ranked lower; one-click "confirm figures"; HMIS sync in phase 2.
Wrong AI understanding	AI only raises urgency (severity = max of AI and rules); deterministic matching; "not a diagnosis" message; Call 108 always visible.
No hospital answers	60-second timeout → next suitable hospital asked automatically.
Poor network	Offline PWA: triage and matching run on the phone from the last saved status.
Privacy / misuse	Consent, masked IDs, hospital-scoped access, OTP login, roles, audit log, 24 h retention.
Fake hospital accounts	Staff accounts cannot be self-registered; issued by the hospital / state.